

Assignment 104

$$P(\text{Detect Lie}) = 0.96$$

$$P(\text{Detect Truth}) = 0.95$$

$$P(\text{Lie}) = 0.2$$

$$P(\text{Lie} | \text{machine dec lie}) = \frac{P(\text{machine dec lie} | \text{Lie}) \cdot P(\text{Lie})}{P(\text{machine dec lie})}$$

$$P(\text{machine dec lie}) = P(\text{machine dec lie} | \text{Lie}) \cdot P(\text{Lie}) + P(\text{machine dec lie} | \text{Truth}) \cdot P(\text{Truth})$$

$$= (0.96 \times 0.2) + (0.05 \times 0.98) \\ = 0.0682$$

$$P(\text{Lie} | \text{machine dec Lie}) = \frac{0.96 \times 0.2}{0.0682} = 0.2816$$

b

$$\begin{aligned} P(\text{Truth} | \text{mac dec lie}) &= 1 - P(\text{Lie} | \text{mac dec lie}) \\ &= 1 - 0.28 \\ &= 0.7 \end{aligned}$$

Question 5

Problem of Bayes's Theorem

(1) is a scalability issue

(2) Dependency among features

So

To solve this we use Naive Bayes.

It assumes if a feature is independent or not and solves the complexity of Bayes.

Data Set

Weather	Temp	Class
Sunny	Hot	NO
overcast	Hot	Yes
Rainy	mild	Yes
Sunny	Mild	NO
Overcast	cool	Yes

Naive Bayes

$$P(\text{Yes}|\text{Weather}) \propto P(\text{Yes}) \cdot P(\text{Overcast}|\text{Yes}) \\ P(\text{Mild}|\text{Yes})$$

Question 6

$P(\text{Play Tennis} = \text{Yes})$ and $P(\text{Play Tennis} = \text{No})$

$$P(\text{Play Tennis} = \text{Yes}) = 6/8 = 0.75$$

$$P(\text{Play Tennis} = \text{No}) = 2/8 = 0.25$$

For 'Yes'

$$P(\text{sunny} | \text{Yes}) = \frac{2}{5} = 0.4$$

$$P(\text{Normal} | \text{Yes}) = \frac{3}{5} = 0.6$$

~~P(Humidity)~~

$$P(\text{mild} | \text{Yes}) = \frac{2}{5} = 0.4$$

$$P(\text{True} | \text{Yes}) = \frac{2}{5} = 0.4$$

$$P(\text{Yes}) = 0.625$$

$$P(\text{Yes} | \text{sunny, mild, Normal, True}) \propto 0.625 \times 0.4$$

$$\times 0.4 \times 0.6 \times 0.4$$

$$= 0.024$$

For 'No'

$$P(\text{sunny} | \text{No}) = \frac{0}{3} = 0$$

$$P(\text{mild} | \text{No}) = \frac{2}{3} = 0.6667$$

$$P(\text{Normal} | \text{No}) = \frac{0}{3} = 0$$

$$P(\text{True} | \text{No}) = \frac{1}{3} = 0.33$$

$$P(\text{No}) = 0.375$$

$$P(\text{NO} | \text{sunny, mild, Normal, True}) \propto 0.375 \times 1 \times 0.6667 \times 0 \times 0.333 = 0$$

$$P(\text{NO}) = 0$$

so, player likely to play tennis.

b

for Yes,

$$P(\text{Yes}) \cdot P(\text{Overcast} | \text{Yes}) \cdot P(\text{Hot} | \text{Yes})$$

$$= 0.75 \times 0.33 \times 0$$

$$= 0$$

for NO,

$$P(\text{NO}) \cdot P(\text{Overcast} | \text{NO}) \cdot P(\text{Hot} | \text{NO})$$

$$= 0.25 \times 0 \times 0.5$$

$$= 0$$

Both '0' not prediction

Question 7

$$P(\text{Yes}) = \frac{Y}{T} = \frac{10}{14}$$

$$P(\text{No}) = \frac{N}{T} = \frac{4}{14}$$

$$\begin{aligned} H(\text{Edible}) &= -\left(\frac{10}{14} \log_2\left(\frac{10}{14}\right) + \frac{4}{14} \log_2\left(\frac{4}{14}\right)\right) \\ &= -(-0.34 + (-0.517)) \\ &= 0.863 \end{aligned}$$

$$P(\text{Yes} | \text{Yellow}) = \frac{7}{10}$$

$$P(\text{No} | \text{Yellow}) = \frac{3}{10}$$

$$\begin{aligned} H(\text{Yellow}) &= -\left(\frac{7}{10} \log_2\left(\frac{7}{10}\right) + \frac{3}{10} \log_2\left(\frac{3}{10}\right)\right) \\ &= 0.88 \end{aligned}$$

For Green,

$$P(\text{Yes}|\text{Green}) = \frac{3}{4} \quad P(\text{No}|\text{Green}) = \frac{1}{4}$$

$$H(G) = -\left(\frac{3}{4} \log_2 \left(\frac{3}{4}\right) + \frac{1}{4} \log_2 \left(\frac{1}{4}\right)\right)$$

$$= 0.811$$

For color

$$H(\text{color}) = \frac{10}{14} \times 0.881 + \frac{4}{14} \times 0.811 = 0.865$$

Info gain,

$$IG(\text{color}) = H(\text{Edible}) - H(\text{color})$$

$$= 0.863 - 0.865$$

$$= -0.002$$

Size

small: 10, large: 5

$$H(\text{small}) = -\left(\frac{7}{10} \log_2 \left(\frac{7}{10}\right) + \frac{3}{10} \log_2 \left(\frac{3}{10}\right)\right)$$

$$= 0.882$$

$$H(\text{large}) = -\left(\frac{3}{5} \log_2 \left(\frac{3}{5}\right) + \frac{2}{5} \log_2 \left(\frac{2}{5}\right)\right)$$

$$= 0.971$$

Weighted Entropy,

$$H(\text{size}) = \frac{10}{15} \times 0.88 + \frac{5}{15} \times 0.97 = 0.912$$

$$IG(\text{size}) = H(S) - H(\text{size})$$

$$= 0.918 - 0.912 = 0.006$$

Shape

$$H(\text{Round}) = - \left(\frac{8}{12} \log_2 \left(\frac{8}{12} \right) + \frac{4}{12} \log_2 \left(\frac{4}{12} \right) \right)$$

$$= 0.918$$

$$H(\text{Irregular}) = - \left(\frac{2}{3} \log_2 \left(\frac{2}{3} \right) + \frac{1}{3} \log_2 \left(\frac{1}{3} \right) \right)$$

$$= 0.918$$

Weighted.

$$H(\text{shape}) = \frac{12}{15} \times 0.918 + \frac{3}{15} \times 0.918$$

$$= 0.918$$

$$IG = H(S) - H(\text{shape}) = 0.918 - 0.918$$

~~shape~~ size

color has the highest Information gain.